

Module 4: Integration

Ph.D. Math Camp

1 Antiderivatives

In many problems, the derivative of a function is known, and the goal is to find the function f itself. For example, a sociologist who knows the rate at which the population is growing may wish to use this information to predict future population levels.

The process of obtaining a function f from its derivative f' is called **antidifferentiation** or **integration**.¹

Consider a function f and suppose it is the derivative of another function, say F . Then, $F'(x) = f(x)$ for every x in the domain of f . $F(x)$ is said to be an antiderivative of $f(x)$.

A function has more than one antiderivative. For example, one antiderivative of the function $f(x) = 3x^2$ is $F(x) = x^3$ since $F'(x) = 3x^2 = f(x)$.

But $x^3 + 12$ and $x^3 + e$ are also antiderivatives of $f(x) = 3x^2$.

Fundamental Property of Antiderivatives If F is an antiderivative of the continuous function f , then any other antiderivative of f differs from F by a constant.

This is because any two functions with the same derivative (slope) in a domain must have parallel graphs (i.e., differ by a constant).

¹Not every function has an antiderivative expressible in elementary terms (e.g., e^{-x^2}). However, most of what we will deal with will have.

We will represent the family of all antiderivatives of $f(x)$ by using the symbolism

$$\int f(x)dx = F(x) + C$$

which we call the **indefinite integral** of f . Thus, $F(x) + C$ has derivative $f(x)$. dx , as we saw before, indicates the antidifferentiation process with respect to the variable x .

Differentiate your answer $F(x) + C$ to check the antidifferentiation calculation. If the derivative equals $f(x)$, your calculation is correct, but you've made a mistake if it is anything other than $f(x)$.

Rules for Integrating Common Functions

- Constant Rule: $\int k dx = kx + C$
- Power Rule: $\int x^n dx = \frac{1}{n+1}x^{n+1} + C$ (for $n \neq -1$)
- Logarithmic Rule: $\int \frac{1}{x} dx = \ln |x| + C$ for all $x \neq 0$
- Exponential Rule: $\int e^{kx} dx = \frac{1}{k}e^{kx} + C$
- Constant multiple Rule: $\int k f(x) dx = k \int f(x) dx$
- Sum Rule: $\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$
- Difference Rule: $\int [f(x) - g(x)] dx = \int f(x) dx - \int g(x) dx$

Examples:

(i) $\int 4 dx$

(ii) $\int 5x^4 dx$

(iii) $\int 9e^{-3x} dx$

(iv) $\int dx$

(v) $\int x^{\frac{2}{3}} dx$

$$\text{(vi)} \int x^{-\frac{1}{5}} dx$$

$$\text{(vii)} \int \frac{dx}{x}$$

$$\text{(viii)} \int \frac{1}{3x} dx$$

$$\text{(ix)} \int \sqrt{x} dx$$

$$\text{(x)} \int \frac{dx}{\sqrt[3]{x}}$$

$$\text{(xi)} \int (5x^3 + 2x^2 + 3x) dx$$

$$\text{(xii)} \int (2x^6 - 3x^4) dx$$

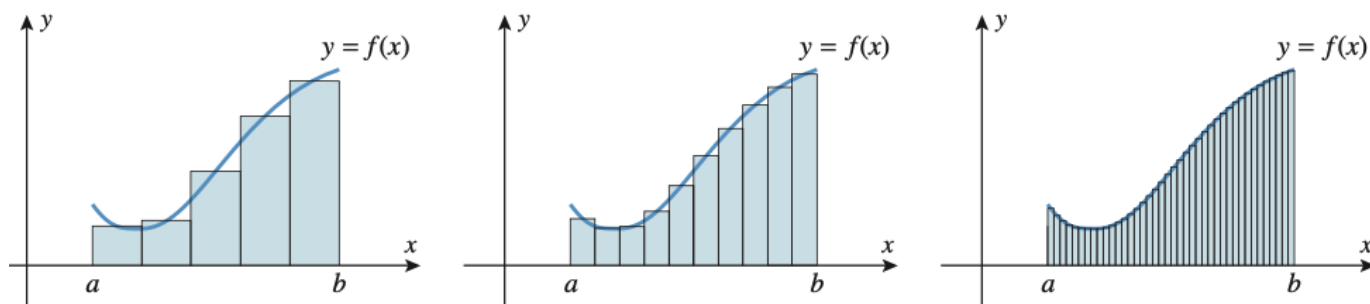
2 Definite Integration

We now revisit the area problem. How do we find the area between the graph of f and the interval $[a, b]$ on the x -axis.

As we did with the tangent line problem, we start with an approximation using what we know.

Divide the interval $[a, b]$ into n equal subintervals, and over each subinterval, construct a rectangle that extends from the x -axis to any point on the curve $y = f(x)$ that is above the subinterval

Figure 1: Rectangle Method



Generally, we define an integral as follows.

Step 1: Subdivide $[a, b]$ into subintervals: $a = x_0 \leq x_1 \leq x_2 \leq \dots \leq x_n = b$. Let

$\Delta x_i = x_i - x_{i-1}$ (the width of the interval)

Step 2: Let $e_i \in [x_{i-1}, x_i]$ e_i is any point in the subinterval.

Step 3: The area of the rectangles can be expressed as

$$f(e_1)\Delta x_1 + f(e_2)\Delta x_2 + \dots + f(e_n)\Delta x_n = \sum_{j=1}^n f(e_j)\Delta x_j$$

Step 4: We can interpret this sum as the integral

$$\int_a^b f(x)dx = \lim_{\max \Delta x \rightarrow 0} \sum_{j=1}^n f(e_j)\Delta x_j$$

This more general version doesn't restrict the rectangles to any particular width.

Δx_j could be of different sizes. Note that as $n \rightarrow \infty$ $\Delta x_j \rightarrow 0$

In a way,

- $\sum_{i=1}^n \longrightarrow \int_{x=a}^b$
- $f(e_j) \longrightarrow f(x)$
- $\Delta x_j \longrightarrow dx$

If $f(x)$ is a continuous function and $f(x) \geq 0$ on the interval $a \leq x \leq b$, then the region under the curve $y = f(x)$ above $a \leq x \leq b$ has the area:

$$A = \int_a^b f(x)dx$$

The Fundamental Theorem of Calculus If $f(x)$ is a continuous function on the interval $a \leq x \leq b$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

where $F(x)$ is any antiderivative of $f(x)$ on $a \leq x \leq b$.

Properties of Definite Integrals

- $\int_a^b f(x) dx = -\int_b^a f(x) dx$
- $\int_a^a f(x) dx = 0$
- $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$

Thus, to calculate a definite integral, we: 1) figure out an antiderivative F , 2) evaluate at b and a , and then subtract $F(a)$ from $F(b)$.

Examples: Evaluate the following definite integrals.

(i) $\int_1^4 10x dx$

(ii) $\int_1^3 (4x^3 + 6x) dx$

(iii) $\int_0^5 (2x^3) dx$

(iv) $\int_1^{64} x^{-\frac{2}{3}} dx$

(v) $\int_1^2 (x^3 + x + 6) dx$

(vi) $\int_0^3 4e^{2x} dx$

(vii) The area of the region under the line $y = 2x + 1$ and above the interval $1 \leq x \leq 3$.

2.1 Integration by Parts

Recall the product rule:

$$(f(x)g(x))' = f'(x)g(x) + g'(x)f(x)$$

Integrate both sides

$$\implies f(x)g(x) = \int f'(x)g(x)dx + \int g'(x)f(x)dx$$

Rearrange,

$$\int g'(x)f(x)dx = f(x)g(x) - \int f'(x)g(x)dx$$

For example, $\int xe^x dx$. Here $f(x) = x, g'(x) = e^x, f'(x) = 1, g(x) = e^x$

$$\implies xe^x - \int 1 \cdot e^x dx = xe^x - e^x + c$$

Examples:

1. $\int \ln x dx$
2. $\int x \ln x dx$
3. $\int x^2 e^{-x} dx$

2.2 Differentiation with Respect to Limits

Suppose a definite integral has a variable upper limit of integration, a constant lower limit of integration, and a continuous integrand. In that case, the derivative of the integral with respect to its upper limit is equal to the integrand evaluated at the upper limit.

$$\frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x)$$

I have used t rather than x as the variable of integration to avoid confusion with the x that appears as the upper limit of integration.

Let

$$A(x) = \int_a^x f(t) dt = F(x) - F(a)$$

$$\implies A'(x) = F'(x) - \cancel{\frac{d}{dx} F(a)}^0$$

$$F'(x) = f(x) \implies A'(x) = f(x)$$

Example:

$$\frac{d}{dx} \left[\int_1^x t^3 dt \right] = x^3$$

We can show this

$$\int_1^x t^3 dt = \frac{x^4}{4} - \frac{1}{4} \implies \frac{d}{dx} \left[\frac{x^4}{4} - \frac{1}{4} \right] = x^3$$

2.3 Consumer Surplus

Willingness-to-Pay

A buyer's maximum amount they would be willing to pay for a good is their willingness to pay; a measure of how much that buyer values the good.

Suppose you go to buy a car that you have decided, beforehand, that the most you will pay is \$20,000. If the car is more than that you do not buy it.

If you find the car at \$15K, then you got a bargain $\rightarrow$ you get a consumer surplus (CS) of 5K. CS is amount a buyer is willing to pay for a good minus the amount buyer actually pays for it.

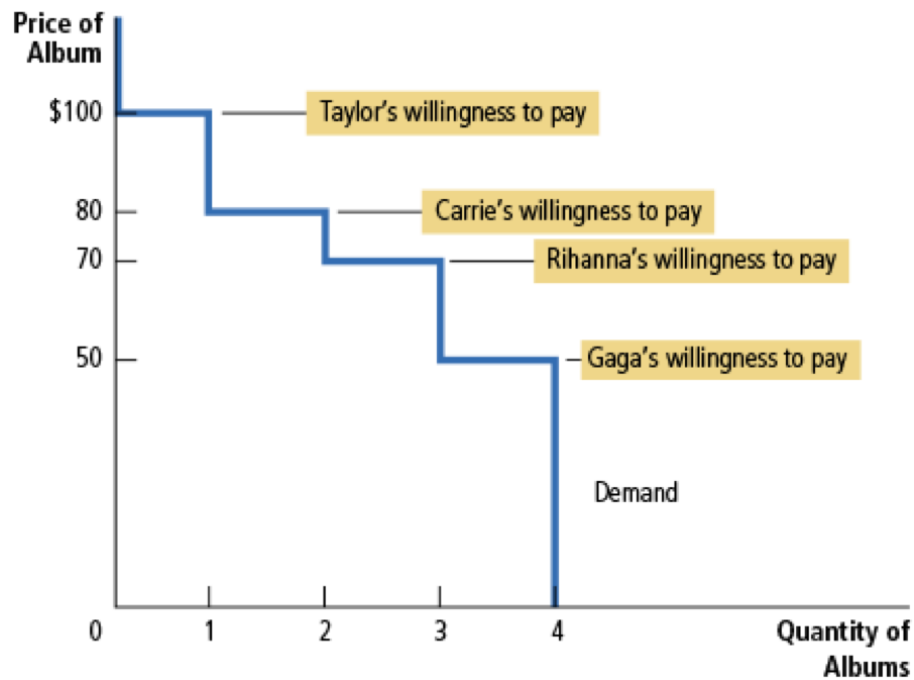
For all demand the area below demand and above price measures CS in a market (i.e. the sum of the individual CS of all buyers).

Using Demand to Measure CS

CS is closely related to the demand curve.

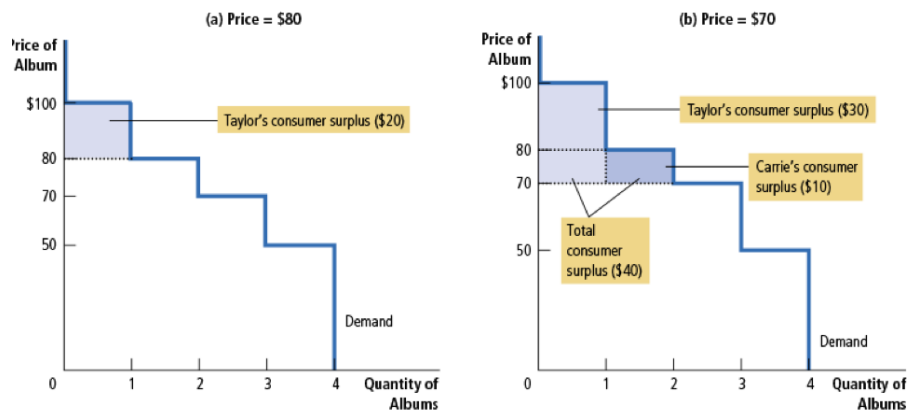
The example from the book is a market for a classic album. Consider the following consumers

Buyer	WTP	P	Q
none	n/a	$P > 100$	0
A	100	80-100	1 (A)
B	80	70-80	2 (A,B)
C	70	50-70	3 (A,B, C)
D	50	$P < 50$	4 (A,B, C, D)



Above 100 nobody consumes. Now that we have constructed the demand curve we can measure the CS.

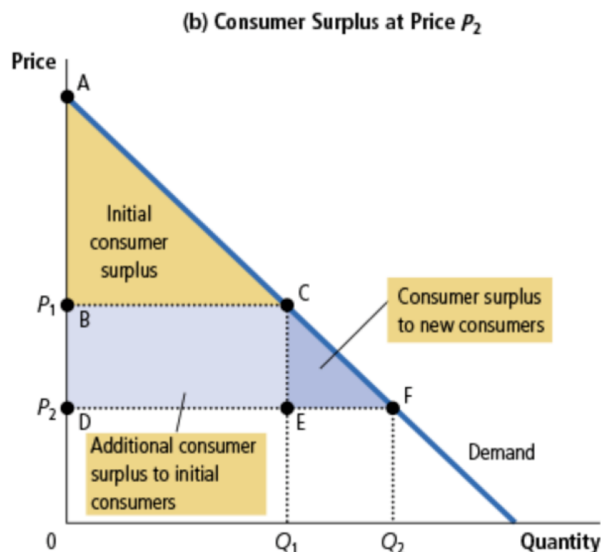
Suppose that $P^* = 80$. Notice that the area above the price and below the **market** demand = 20. This is the CS of person A.



Suppose that $P^* = 70$. Notice that the area above the price and below the **market** demand = 40. This is the CS of person A + CS of B.

Lower price and CS

As $P \downarrow$ consumers are better off. We can use CS to see how well-being is affected.



If we lower the price quantity increased and there is additional CS to original consumers (BCED) and to new consumers (CEF). New total CS AFD.

What does CS Measure?

CS is a pretty good measure of economic well-being because it is the benefits buyers receive from good as the buyers perceive them.

In some instances, we may choose to ignore CS. For instance, the market for heroin.

Measuring CS becomes particularly important when considering how different market structures affect consumer well-being.

Similarly, **Producer Surplus (PS)** is the amount a seller is paid minus the cost of production– benefit to sellers. If the cost is \$ 500 (willingness to sell) but the seller does the job for \$ 600 the PS is \$100.

Example: Consider the following market demand $Q_D = (10 - P)^{\frac{1}{2}}$ and market supply expressed as $Q_S = \frac{1}{3}P$.

1. Find the equilibrium price and quantity. Find the consumer and producer surplus at the equilibrium price and quantity.

Rewrite Q_D, Q_S as functions of price and set equal to each other: $P = 10 - Q^2$ and $P = 3Q$

$$10 - Q^2 = 3Q \implies Q^2 + 3Q - 10 = 0 \implies (Q + 5)(Q - 2) = 0 \implies Q = 2, Q = -5$$

Since prices and quantities cannot be negative, we will stick with $Q = 2 \implies P = 6$.

Note that the demand has a y -intercept at $P = 10$. To find the consumer surplus, we will need to find the area under the demand but above the equilibrium price.

First, find the total area under demand up to $Q = 2$.

$$\int_0^2 (10 - Q^2) dQ = 10Q \Big|_{Q=0}^2 - \frac{Q^3}{3} \Big|_{Q=0}^2 = 20 - 0 - \frac{8}{3} + 0 = \frac{52}{3}$$

Subtract cost to consumer which is area under the price to $q = 2$. That is just 12.

$$\text{Thus } CS = \frac{52}{3} - \frac{36}{3} = \frac{16}{3}$$

For the producer surplus, take the revenue received at the equilibrium ($P^*Q^* = 12$) and subtract the area under the supply curve up to $Q = 2$:

$$PS = \int_0^2 6 \, dQ - \int_0^2 3Q \, dQ = 6Q - \frac{3}{2}Q^2 \Big|_{Q=0}^2 = 12 - 6 = 6$$